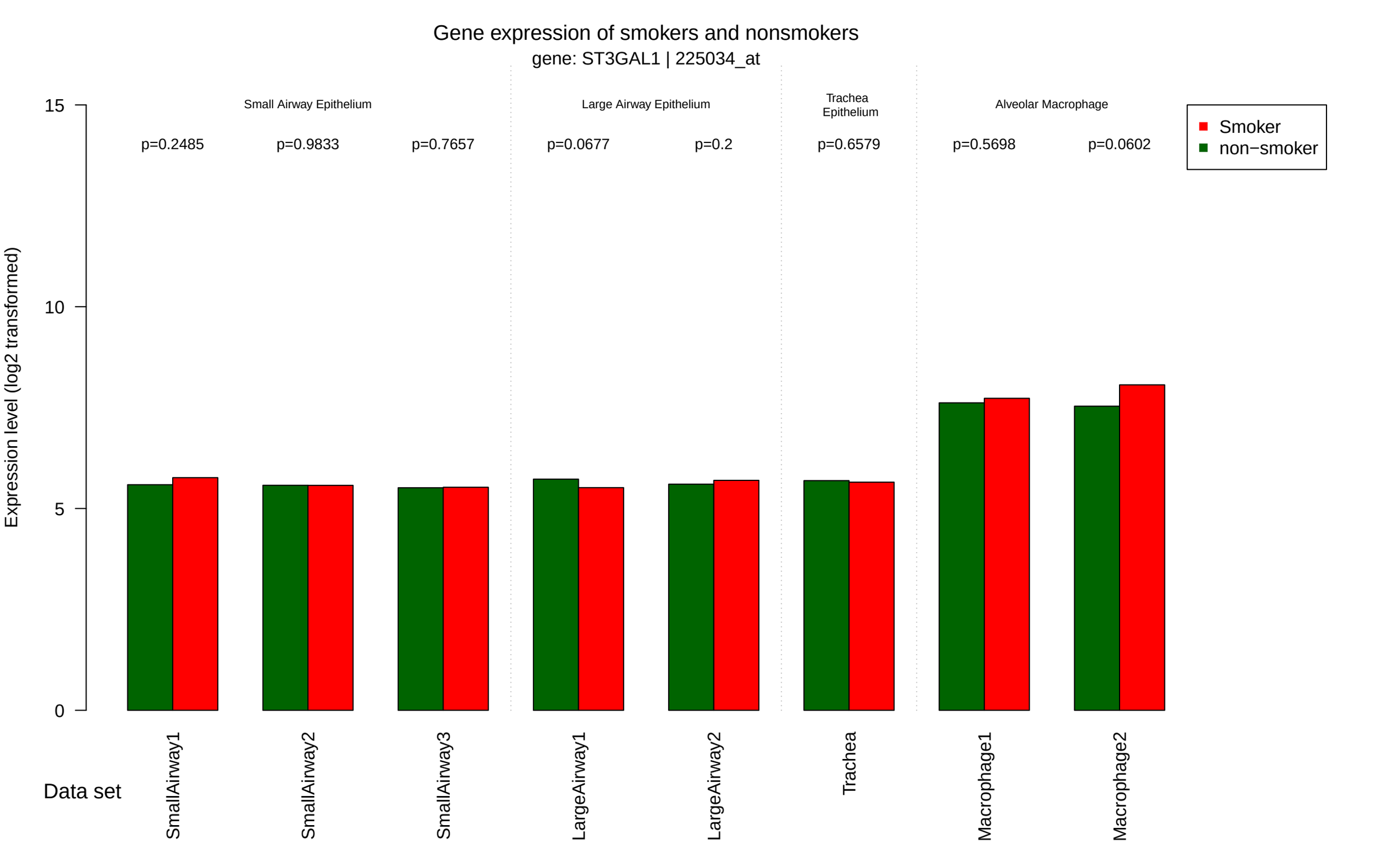
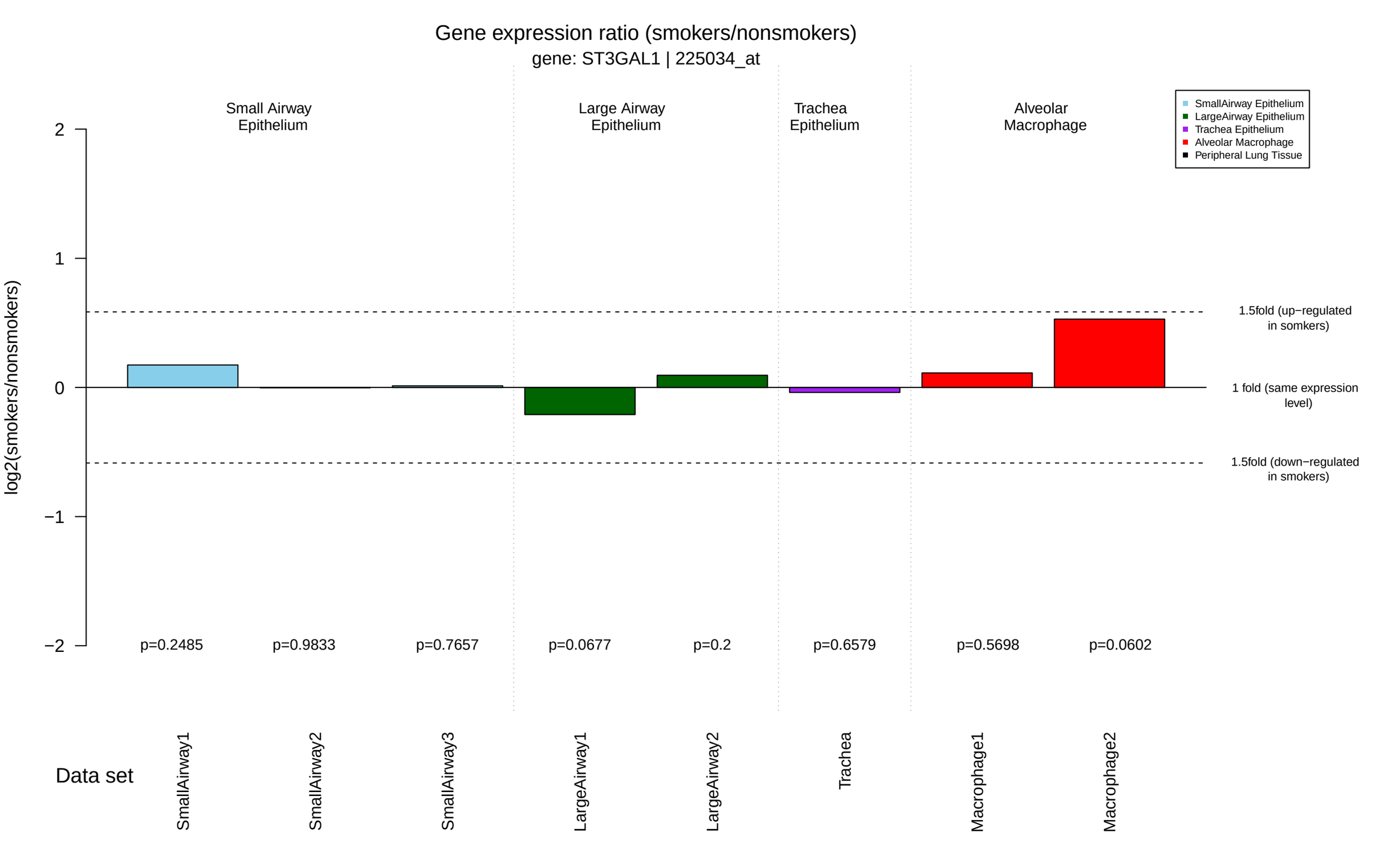
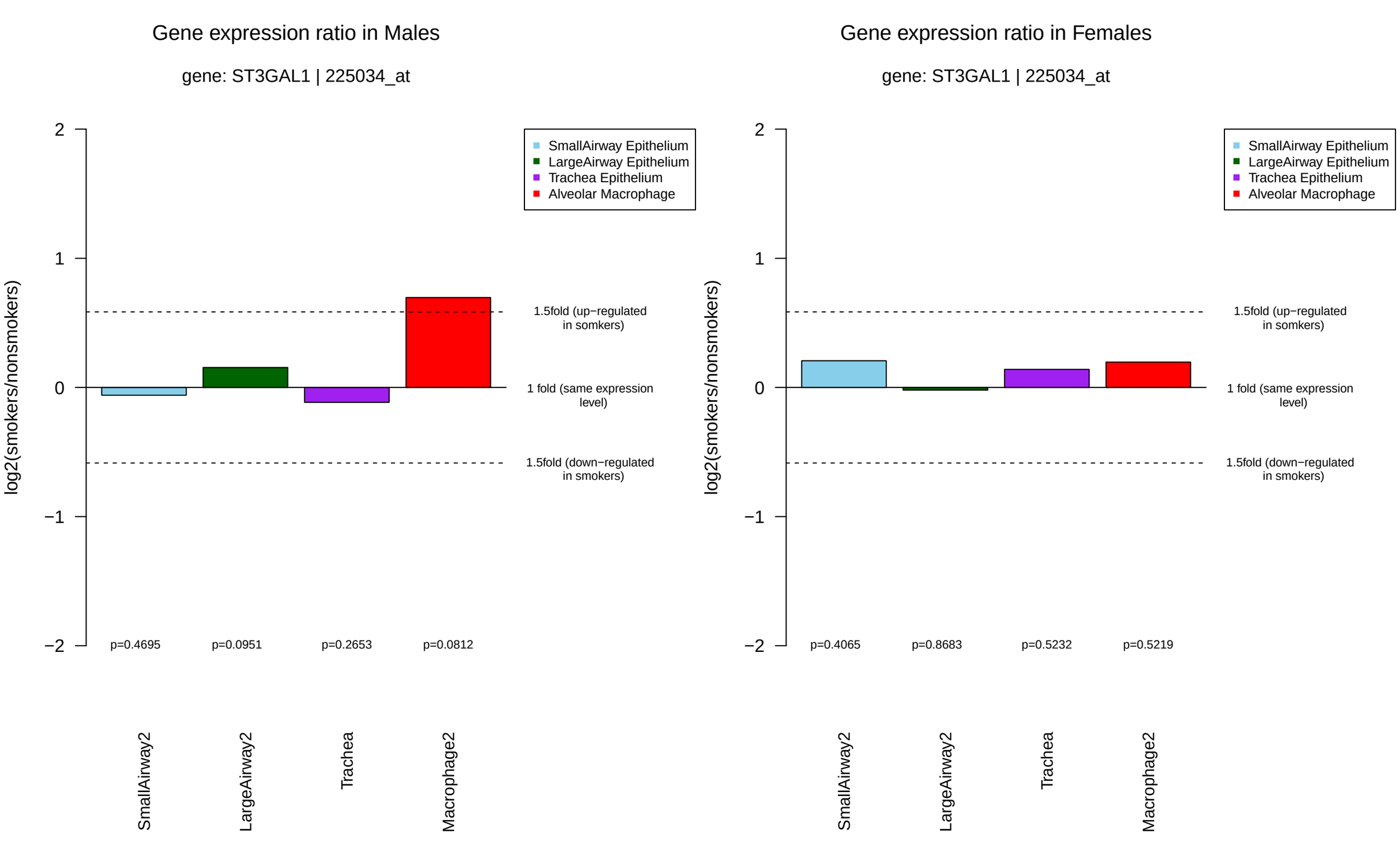
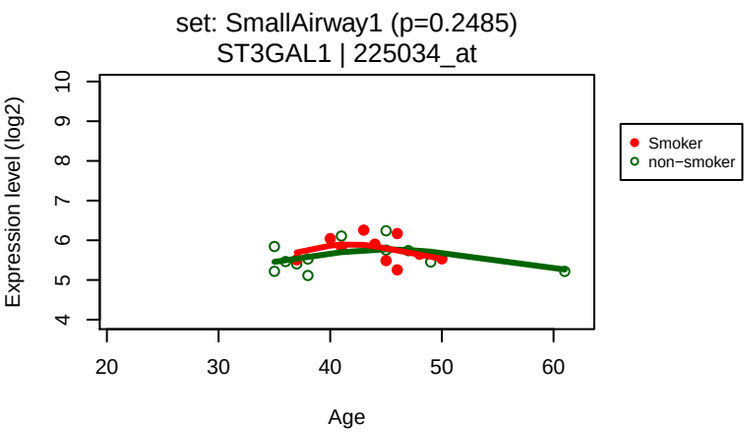
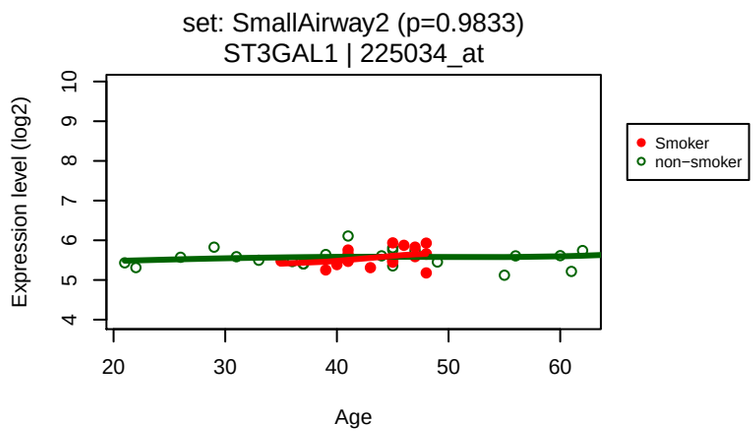
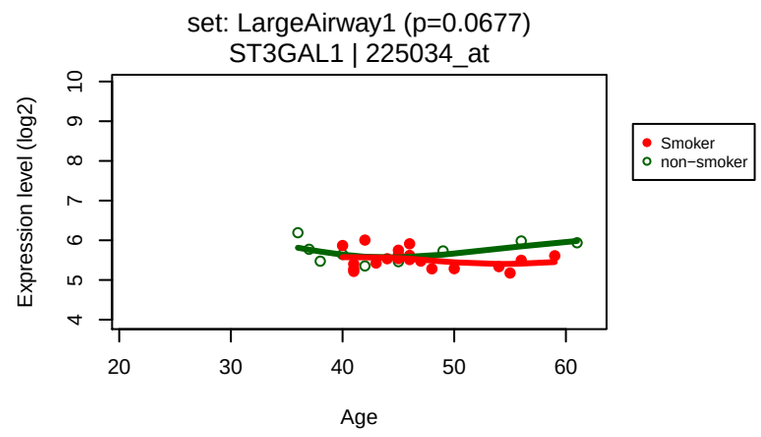

Gene expression of smokers and nonsmokers
gene: ST3GAL1 | 225034_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2485
p=0.9833
p=0.7657
p=0.0677
p=0.2
p=0.6579
p=0.5698
p=0.0602
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

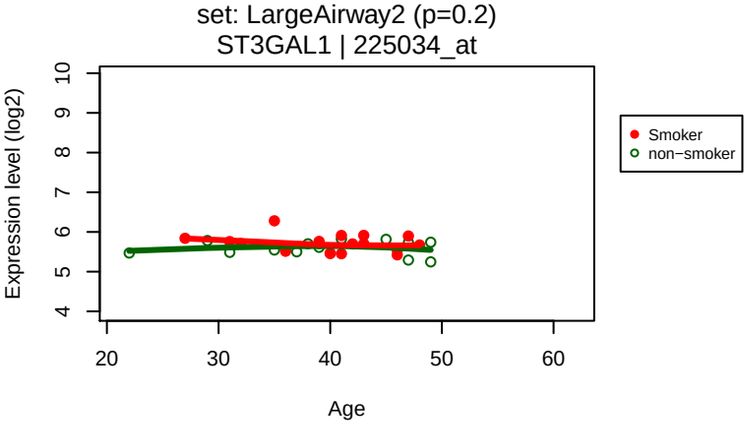

Gene expression ratio (smokers/nonsmokers)
gene: ST3GAL1 | 225034_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2485
p=0.9833
p=0.7657
p=0.0677
p=0.2
p=0.6579
p=0.5698
p=0.0602
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

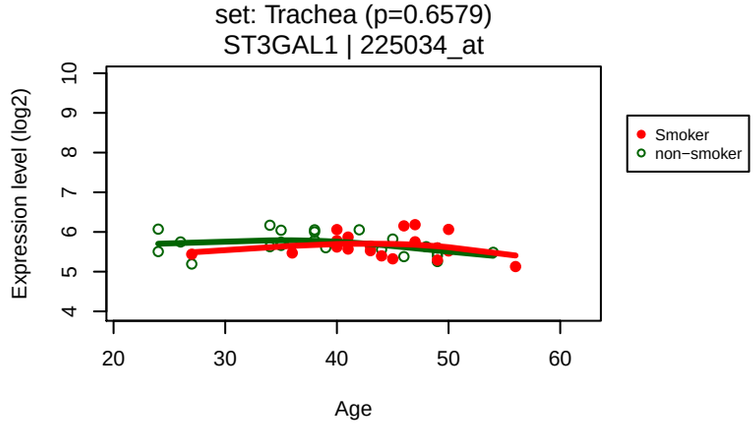

Gene expression ratio in Males
gene: ST3GAL1 | 225034_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4695
p=0.0951
p=0.2653
p=0.0812
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST3GAL1 | 225034_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4065
p=0.8683
p=0.5232
p=0.5219
SmallAirway2
LargeAirway2
Trachea
Macrophage2

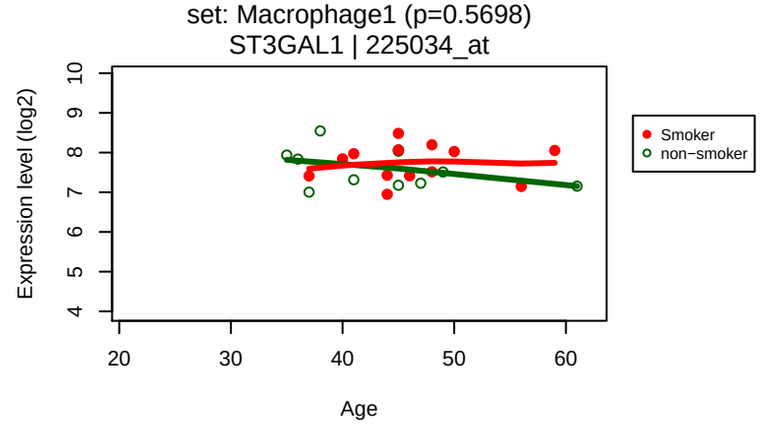

set: SmallAirway1 (p=0.2485)
ST3GAL1 | 225034_at
Expression level (log2)
Age
Smoker
non−smoker

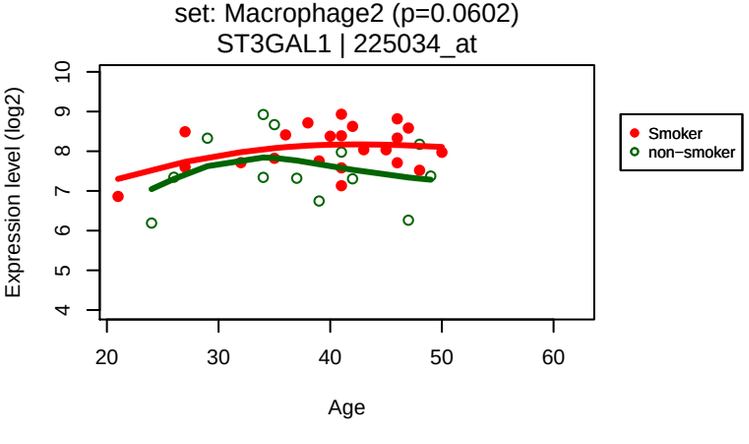

set: SmallAirway2 (p=0.9833)
ST3GAL1 | 225034_at
Expression level (log2)
Age
Smoker
non−smoker

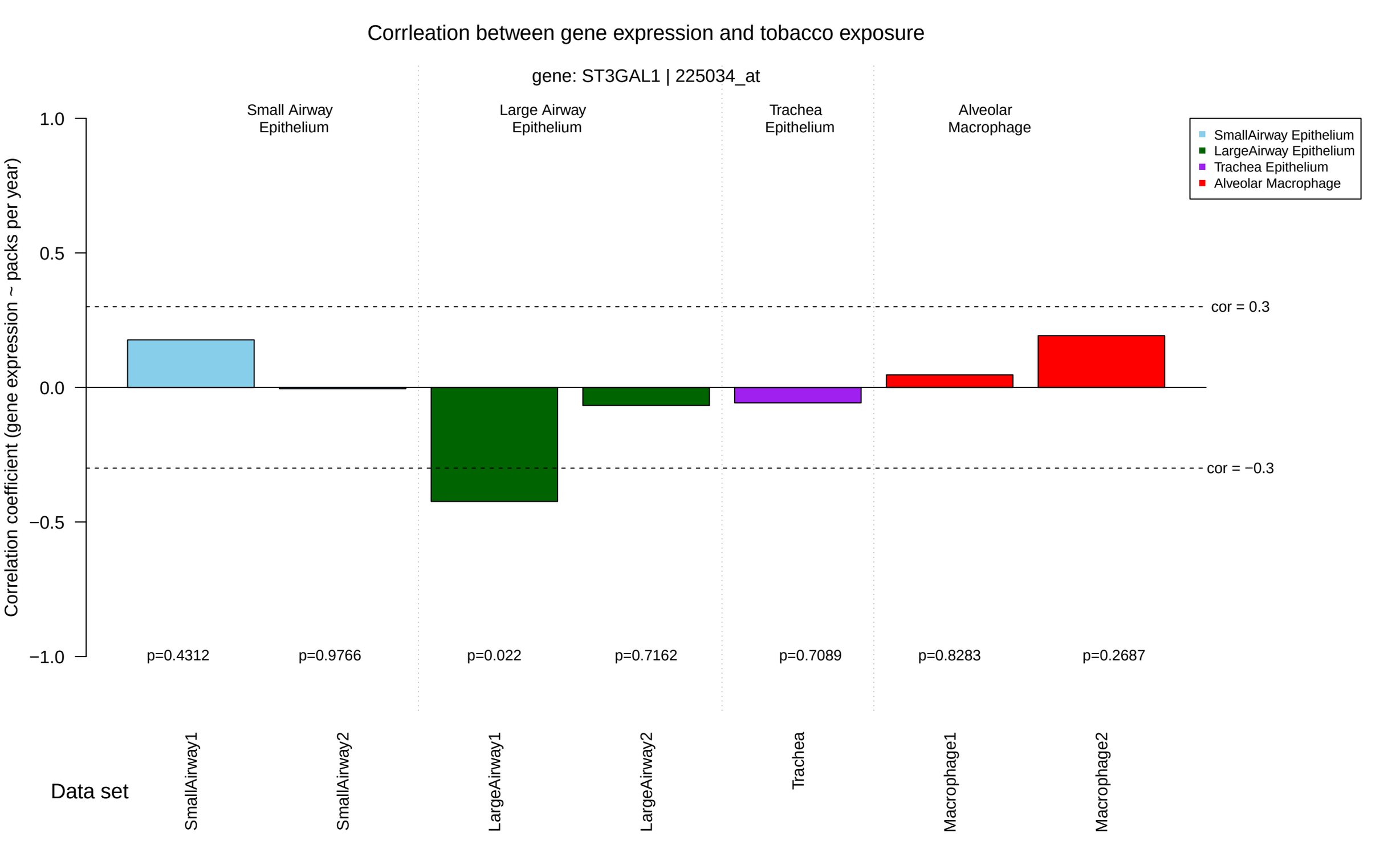

set: LargeAirway1 (p=0.0677)
ST3GAL1 | 225034_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2)
ST3GAL1 | 225034_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6579)
ST3GAL1 | 225034_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5698)
ST3GAL1 | 225034_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0602)
ST3GAL1 | 225034_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST3GAL1 | 225034_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4312
p=0.9766
p=0.022
p=0.7162
p=0.7089
p=0.8283
p=0.2687
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2